Verified Self Training in a Small Language Model

A prespecified case study on GSM Symbolic

Lewis Adkins

Abstract. We tested single-round verified self-training with LoRA on Qwen3-1.7B-Base, using GSM8K training problems and 400 GSM-Symbolic development questions from 37 template families. Four sequentially prespecified primary tests covered a 43-example pilot, difficulty targeting, human-reference supervision, and training on 2,000 verified examples. None met its primary criterion. The largest self-training condition gained 1.58 percentage points over base, with a 95% seed-and-family interval of [−3.08, +6.50]. Human-reference supervision, intended as a positive control, changed accuracy by −6.33 points, with a family-only interval of [−14.52, +2.28]. Without a control that demonstrated improvement, the experiments leave recipe effectiveness and pipeline sensitivity unresolved. A base-model check changed accuracy by 1.00 point when only inference batch size changed. We document the case with three-seed results, fixed decision rules, completion diagnostics, and scoring checks.

1 Introduction

Verified self-training uses a model's correct outputs as fine-tuning targets. It offers a practical way to study adaptation with limited compute: generate candidate solutions, check their answers, and train on the retained examples. The question here is whether a minimal version produces a detectable gain on a small base model under a fixed evaluation protocol.

We trained Qwen3-1.7B-Base adapters on verified GSM8K solutions and evaluated them on template-generated GSM-Symbolic questions. The conditions varied which problems supplied the targets and how much training was performed. A human-reference condition used the same hard problems as a supervised control. Success criteria were recorded before the corresponding evaluations, after earlier development results had been inspected. These are sequentially prespecified development comparisons, not a publicly registered or held-out confirmatory study.

The contribution is a documented case study: this minimal recipe and its intended positive control both failed to show a reliable improvement under their primary criteria. We report all primary and secondary contrasts, including secondary intervals that exclude zero, and distinguish original decisions from later sensitivity analyses. As an evaluation baseline, changing only the base model's inference batch size moved accuracy from 39.25% to 38.25%.

2 Related work

STaR generates rationales, retries unsuccessful problems with answer hints, and repeats fine-tuning [Zelikman et al., 2022]. Rejection-sampling fine-tuning (RFT) retains distinct correct reasoning paths; its reference experiments sample 100 candidates per problem [Yuan et al., 2023, Section 3.3 and Appendix A.3]. ReST-EM alternates generation, filtering, and training, sampling 32 candidates per MATH problem at each iteration and retaining up to 10 correct solutions per problem for fine-tuning [Singh et al., 2023, Section 5]. Our recipe uses one round, four candidates, and one retained solution per problem, so it is smaller in sampling, iteration count, and retained targets.

Prior work studies familiarity with generated targets [Ren et al., 2024], self-distillation to reduce distribution gaps [Yang et al., 2024], and variance in benchmark evaluation [Madaan et al., 2024]. GSM-Symbolic introduces controlled numerical and clause variations [Mirzadeh et al., 2025]. These works motivate the training comparisons and diagnostics in this case study.

3 Methods

3.1 Model and training data

We fine-tune LoRA adapters [Hu et al., 2022] on Qwen3-1.7B-Base [Yang et al., 2025], keeping the base weights frozen. Later experiments use snapshot ea980cb0a6c2ae4b936e82123acc929f1cec04c1, with a verified weight checksum. The original pilot did not record a model commit. All training used one NVIDIA RTX 2080 Super with 8 GB of memory.

GSM8K training problems [Cobbe et al., 2021] were shuffled with seed 2026. All positions below refer to that shuffled order. The pilot sampled two candidates per problem at positions 222–271 and retained the first correct candidate for each of 43 eligible problems. Later labeling sampled four candidates at temperature 0.7 and top-p 0.9, with a 384-token limit. A problem was hard if 1–3 candidates were correct and easy if all four were correct. One correct candidate was chosen uniformly per eligible problem. The 1,100-problem pool at positions 536–1635 contained 86 problems with no correct attempt, 439 hard, and 575 easy.

Table 1 Training sets and update counts

Condition	Examples	Supervised tokens per epoch	Updates per seed
Pilot	43	Not logged	22 recorded
Hard	300	32,879	150
Random	353	32,619	178
Easy	403	32,592	202
Human references	300	32,206	150
Scaled random	2,000	184,572	1,000

All configurations used seeds 2026, 2027, and 2028; two epochs; LoRA rank 4 and alpha 8 on query and value projections; AdamW with constant learning rate 1e-4 and no weight decay; effective batch 4; gradient clipping 1.0; and FP16. Prompt tokens were masked, sequence length was limited to 1,024, and the final adapter was retained. Appendix A gives the prompt and full settings.

3.2 Evaluation and uncertainty

The common evaluation has 200 P1 and 200 P2 questions: 20 templates per variant, with instances 5–14. They span 37 underlying families; 63 other families remain reserved and unevaluated. P1 and P2 add one and two clauses. Greedy generation uses batch size 8 and at most 384 new tokens. The primary scorer extracts a leading numeric fragment after #### and compares saved Decimal representations. Scoring limitations and sensitivity checks are reported in Appendix F.

We use paired percentile bootstrap intervals with 10,000 replicates and random seed 2026. Each draw resamples 37 families with replacement and retains their questions together. Shared family weights apply to all models and both variants; accuracy uses weighted correct totals divided by weighted question totals. Three-seed means use equal seed weights. Family-only intervals hold the trained adapters fixed. Seed-and-family intervals first resample three paired seed indices and then one shared family draw. The pooled scaled-minus-base interval was prespecified; pooled scaled-minus-random was a prospectively specified optional sensitivity. All other seed-and-family intervals here are post hoc. All secondary intervals are marginal and unadjusted for multiple comparisons.

4 Results

None of the four primary tests met its prespecified criterion. The intended human-reference positive control also failed to show improvement, so this case leaves the recipe's effectiveness and the pipeline's sensitivity unresolved. Table 2 reports all pooled contrasts, with the original family-only results and seed-and-family sensitivities presented consistently.

Table 2 Pooled three seed comparisons in percentage points

Comparison	Role	Change	95% family interval	95% seed and family interval
Pilot 43 − Base	T1	+3.42	[−1.26, +8.00]	[−1.25, +8.25]
Hard 300 − Random 353	T2	−0.92	[−5.00, +2.87]	[−5.85, +3.33]
References 300 − Base	T3	−6.33	[−14.52, +2.28]	[−15.25, +3.33]
Scaled 2000 − Base	T4	+1.58	[−2.14, +5.50]	[−3.08, +6.50]
Hard 300 − Base	S	+2.00	[−2.31, +6.34]	[−2.50, +6.43]
Random 353 − Base	S	+2.92	[−1.11, +7.07]	[−1.28, +7.24]
Easy 403 − Base	S	−0.92	[−5.95, +3.92]	[−6.35, +4.50]
Hard 300 − Easy 403	S	+2.92	[−1.54, +7.50]	[−2.31, +8.95]
Random 353 − Easy 403	S	+3.83	[−1.58, +10.09]	[−2.14, +10.43]
References 300 − Hard 300	S	−8.33	[−16.50, −0.25]	[−16.90, +0.81]
Scaled 2000 − Random 353	S	−1.33	[−5.12, +2.72]	[−6.25, +3.41]

T1–T4 identify the four primary tests; S denotes a prespecified secondary contrast. Seed-and-family intervals are post hoc except pooled scaled minus base (prespecified) and pooled scaled minus random-353 (prospectively specified optional sensitivity). New sensitivities reuse saved verdicts and do not replace original decisions. Appendix B gives exact rules; Appendix D reports variant contrasts.

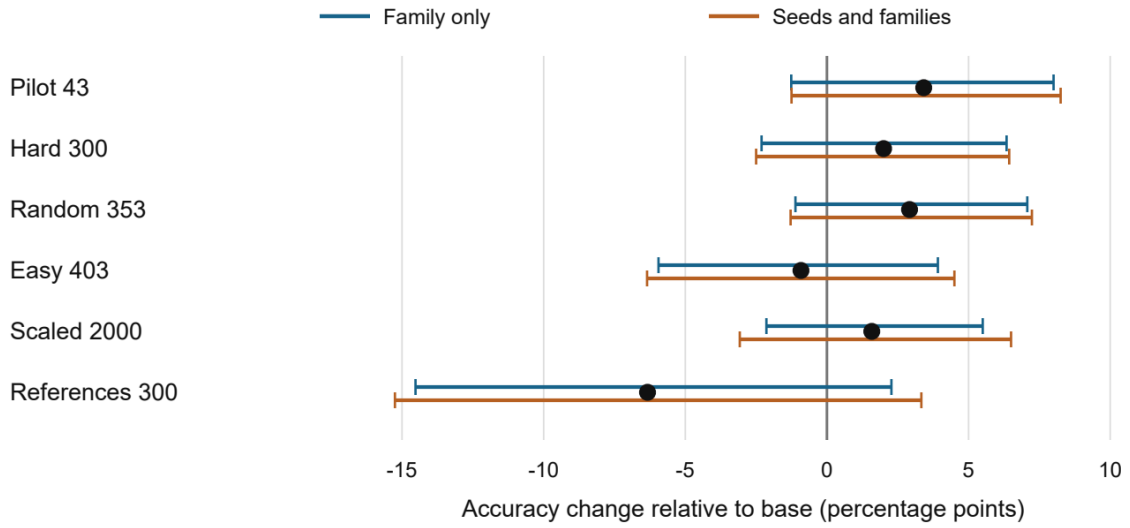


Figure 1. Three-seed mean changes relative to base on the same 400 questions. Blue bars resample families; orange bars resample seeds and families. The latter are post hoc except for scaled-2,000. The plotted intervals describe differences in accuracy, not uncertainty in individual seed scores.

4.1 Pilot and difficulty targeting

The 43-example pilot first scored 40.00% against base's 37.50% on instances 0–4, a 2.50-point exploratory gain. After the expanded-evaluation rule was recorded, the same adapters scored a mean 42.67% on instances 5–14, against 39.25% for base: +3.42 points [−1.26, +8.00]. The primary benefit criterion required a strictly positive family-only lower endpoint, with a mandatory numeric-scoring check for formatting robustness. Numeric rescoring changed no verdicts; the result remained inconclusive.

For difficulty targeting, supervised-token matching yielded 300 hard, 353 random, and 403 easy examples. Their mean accuracies were 41.25%, 42.17%, and 38.33%. The primary hard-minus-random contrast was −0.92 points [−5.00, +2.87]. Its rule favored hard for an interval strictly above zero and random for an interval strictly below zero; otherwise the result was inconclusive. Matching tokens left different update counts: easy received 34.67% more updates than hard. Hard/random overlapped on 101 problems and random/easy on 140.

All three pooled condition-versus-base family intervals included zero. On P1 alone, random minus base was +4.67 points [+0.14, +9.63], a borderline unadjusted secondary result among multiple comparisons. It did not replace the pooled primary decision. Appendix D reports every P1/P2 contrast, including the P2 hard-minus-easy interval whose lower endpoint is exactly zero.

4.2 Human reference control

The reference condition replaced generated targets with GSM8K reference solutions for the same 300 hard problems, in the same order. It used 150 updates per seed and 32,206 supervised tokens per epoch, 2.05% fewer than hard. Mean accuracy was 32.92%, a change of −6.33 points [−14.52, +2.28] relative to base. The prespecified control criterion required a strictly positive family-only lower endpoint. The outcome was inconclusive. Calculator annotations of the form <<...=...>> were stripped from the reference targets; the stored wording and final numeric answers were otherwise retained.

References minus hard was a prespecified secondary contrast, with no separate efficacy gate. Its family-only interval was −8.33 points [−16.50, −0.25], favoring hard conditional on the observed adapters. The post hoc seed-and-family interval was [−16.90, +0.81]. We report both: the first excludes zero without multiplicity adjustment, while the second includes zero after accounting for the observed training-seed variation.

Reference training loss in epoch two was about 0.48, versus about 0.16 for generated hard targets. Mean response lengths were 174.49 tokens for references, 174.32 for base, and 167.44 for hard. These similar averages coexist with different completion behavior. Base reached the 384-token cap on 20/400 responses (5.00%); hard did so on 35/1,200 (2.92%); references on 52/1,200 (4.33%). Of those cap hits, 12, 31, and 49 respectively were unparsed. Thus reference models hit the cap more often than hard, but their pooled cap rate was below base's.

Reference models produced 20, 13, and 16 unparsed answers across the three seeds, compared with 12 for base. They matched base's non-null parsed answer on 321/1,200 responses (26.75%), compared with 1,657/3,600 (46.03%) across hard, random, and easy combined. These agreement rates use all responses as the denominator, rather than conditioning on both answers being parsed. Most reference errors were still parsed: 756 of 805 incorrect responses. Figure 2 and Appendix E give the completion diagnostics.

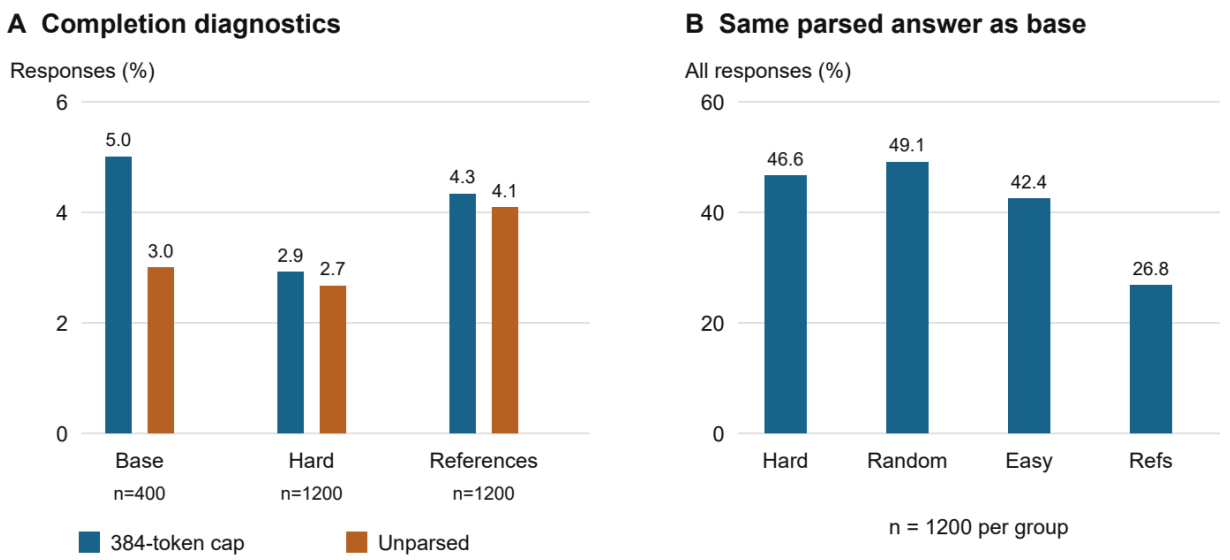


Figure 2. A: cap and unparsed rates use 400 base responses or 1,200 repeated-question responses per trained group. B: agreement means an identical non-null saved prediction, including matches where both models are wrong, divided by all 1,200 responses per group. Repeated responses across seeds are not independent questions. Rounding is for display only.

4.3 Larger random training set

The scaled experiment labeled 2,400 fresh training problems at positions 1636–4035. Of these, 2,244 had a correct candidate; 2,000 eligible problems were selected at random. Each seed received 184,572 supervised tokens per epoch and 1,000 updates. Success required strictly positive lower endpoints for both the family-only and seed-and-family intervals.

Mean accuracy was 40.83%, a gain of +1.58 points. The family-only interval was [−2.14, +5.50] and the seed-and-family interval [−3.08, +6.50]. Relative to random-353, the mean changed by −1.33 points: family-only [−5.12, +2.72], seed-and-family [−6.25, +3.41]. The primary success criterion was not met, and the secondary comparison with random-353 remained inconclusive. The conditions vary in examples, tokens, and updates together; their ordering is not an estimated scaling law.

5 Variation across training and evaluation

The scaled seeds scored 43.50%, 40.75%, and 38.25%, a 5.25-point spread. Depending on the seed, a single run would show a gain of 4.25 points or a loss of 1.00 point relative to base. Reference seeds ranged from 28.75% to 35.75%. Appendix C reports every seed separately.

Across the 12 hard, random, easy, and scaled runs, adapters changed 54–79 of the 400 correctness verdicts relative to base, with net changes from −13 to +17 questions. Including the three pilot runs expands the flip range to 54–83. These counts show how small net changes can accompany many corrections and regressions. They are paired descriptive counts, not independent replications.

The base-only batch-size check used the same 400 questions with batch size 4 instead of 8. It changed 25 final answers and 6 correctness verdicts: one correction and five regressions. Accuracy fell from 39.25% to 38.25%. The study's primary comparisons all retain the saved batch-8 baseline. This one check is an observed evaluation baseline, not a universal threshold below which gains are meaningless.

6 Limitations

The main limitation is the unsuccessful positive control. Without a training condition that produced a detectable gain, the study cannot separate an ineffective minimal recipe from limitations in optimization or evaluation sensitivity. Generation limits are one concrete concern: 49 reference outputs were both capped and unparsed, and each reference seed had 13–20 unparsed answers. Yet most reference errors were parsed, and its cap rate was lower than base's; extending generation alone is not an established remedy.

An earlier GSM8K train-development check scored 174/200 (87%) for the base model, leaving limited headroom on that subset. This was a development check on GSM8K training problems, not a GSM8K test result or a bound on GSM-Symbolic accuracy.

The design covers one small base model, one round of training, four sampled candidates per problem, one retained target per problem, and rank-4 adapters on two projection types. Token matching left unequal update counts, and the scaled experiment increased several quantities together. These choices limit comparisons with richer self-training recipes. All reported efficacy experiments used this single training hyperparameter configuration, without a hyperparameter sweep.

All experiments reuse the same 37 development families, with outputs inspected between experiments. Dated rules constrain subsequent choices but do not create a held-out test. Only three observed training seeds support the seed sensitivities. Percentile intervals over 37 clusters may have imperfect coverage, and secondary intervals are unadjusted. Interval widths describe observed precision rather than a formal power calculation.

Answer-only verification admits flawed reasoning. A targeted review of 20 hard targets flagged two substantive cancelling-error explanations and one flawed, borderline bookkeeping explanation; the author confirmed all three flags on October 14, 2026. This was not a random sample. The legacy scorer also accepts some malformed answer fragments. Separate numeric-equality audits and a stricter scorer for newly generated scaled outputs document these issues without replacing the original primary scores. Original pilot metrics omit a model commit, supervised-token totals, and reliable successful-versus-skipped update counts.

7 Conclusion

This case study found no reliable improvement under four prespecified primary criteria. The largest self-training condition gained 1.58 points on average, while the intended reference control declined by 6.33 points; both primary intervals included zero. The secondary reference-versus-hard family interval favored hard, and a P1 random-versus-base interval narrowly favored random training. Both results remain visible alongside their unadjusted status and the seed sensitivities.

The next step is to establish a training condition with a detectable gain under a fixed scoring and generation protocol. Only then should the 63 reserved families be used for a prespecified confirmatory evaluation.

AI assistance

Claude (Anthropic) and Codex (OpenAI) assisted with code, analyses, literature review, figures, tables, and manuscript drafting and revision. The author designed the study, selected the experiments and success rules, and reviewed the flagged training solutions. The author is responsible for the results, citations, and final text.

Data and code availability

Code, dated plans, aggregate results, and GSM8K-based training data are available at https://github.com/LewisAdk/qwen3-verified-self-training. GSM-Symbolic question text and per-question outputs are not redistributed because of the dataset's CC BY-NC-ND 4.0 licence; LoRA adapter weights are not included. The appendices provide the evaluation prompt, settings, decision rules, per-seed results, secondary contrasts, and diagnostic summaries.

References

Cobbe, K., et al. 2021. Training Verifiers to Solve Math Word Problems. arXiv:2110.14168. https://arxiv.org/abs/2110.14168

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. 2022. LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022. https://arxiv.org/abs/2106.09685

Madaan, L., Singh, A. K., Schaeffer, R., Poulton, A., Koyejo, S., Stenetorp, P., Narang, S., and Hupkes, D. 2024. Quantifying Variance in Evaluation Benchmarks. arXiv:2406.10229. https://arxiv.org/abs/2406.10229

Mirzadeh, S. I., Alizadeh, K., Shahrokhi, H., Tuzel, O., Bengio, S., and Farajtabar, M. 2025. GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models. ICLR 2025. https://openreview.net/forum?id=AjXkRZIvjB

Ren, X., Wu, B., and Liu, L. 2024. I Learn Better If You Speak My Language: Understanding the Superior Performance of Fine-Tuning Large Language Models with LLM-Generated Responses. EMNLP 2024, pp. 10225–10245. https://doi.org/10.18653/v1/2024.emnlp-main.571

Singh, A., Co-Reyes, J. D., Agarwal, R., et al. 2023. Beyond Human Data: Scaling Self-Training for Problem-Solving with Language Models. arXiv:2312.06585. https://arxiv.org/abs/2312.06585

Yang, A., et al. 2025. Qwen3 Technical Report. arXiv:2505.09388. https://arxiv.org/abs/2505.09388

Yang, Z., Pang, T., Feng, H., Wang, H., Chen, W., Zhu, M., and Liu, Q. 2024. Self-Distillation Bridges Distribution Gap in Language Model Fine-Tuning. ACL 2024, Volume 1, pp. 1028–1043. https://doi.org/10.18653/v1/2024.acl-long.58

Yuan, Z., Yuan, H., Li, C., Dong, G., Lu, K., Tan, C., Zhou, C., and Zhou, J. 2023. Scaling Relationship on Learning Mathematical Reasoning with Large Language Models. arXiv:2308.01825. https://arxiv.org/abs/2308.01825

Zelikman, E., Wu, Y., Mu, J., and Goodman, N. D. 2022. STaR: Bootstrapping Reasoning With Reasoning. NeurIPS 2022. https://arxiv.org/abs/2203.14465

Appendix A Prompt and settings

The evaluation and later labeling use the following two-example prefix. The evaluated question is appended as “Question: …” followed by “Solution:”.

Solve the math problems step by step. Finish each solution with #### followed by the numeric answer.

Question: Mitch has saved $20000 to finance his boating hobby. A new boat costs $1500 per foot in length. If Mitch needs to keep $500 for a license and registration, and three times that amount for docking fees, how many feet is the longest boat he can buy?
Solution: Mitch needs to keep 500 * 3 = 1500 for docking fees.
Thus, he needs to keep 500 + 3 * 500 = $2000 for license, registration, and docking fees.
He has 20000 - 2000 = $18000 left to spend on a boat.
Therefore, the longest boat he can buy is 18000 / 1500 = 12 feet long.
12

Question: Stephen rides his bicycle to church. During the first third of his trip, he travels at a speed of 16 miles per hour. During the second third of his trip, riding uphill, he travels a speed of 12 miles per hour. During the last third of his trip, he rides downhill at a speed of 20 miles per hour. If each third of his trip takes 15 minutes, what is the distance Stephen rides his bicycle to church, in miles?
Solution: 15 minutes is 15/60=0.25 hours.
Traveling 15 minutes at 16 miles per hour, Stephen travels 16*0.25 = 4 miles.
Traveling 15 minutes at 12 miles per hour, Stephen travels 12*0.25 = 3 miles.
Traveling 15 minutes at 20 miles per hour, Stephen travels 20*0.25 = 5 miles.
All together, Stephen travels 4+3+5=12 miles.
12

Table A1 Reproduction settings

Setting	Value
Base model	Qwen/Qwen3-1.7B-Base
Later model revision	ea980cb0a6c2ae4b936e82123acc929f1cec04c1
GSM8K revision	740312add88f781978c0658806c59bc2815b9866
GSM Symbolic revision	93b5b3758d9d9841ffe81d6cd2ae2b030685b078
Evaluation	FP16, left padding, greedy, seed 2026, batch 8, 384 new tokens
Candidate generation	Temperature 0.7, top-p 0.9; four candidates; 384 new tokens
Pilot generation	Two candidates, 256 new tokens; retain first correct candidate
LoRA	Rank 4, alpha 8, no dropout or bias; q_proj and v_proj
Optimizer	AdamW, learning rate 1e-4, no decay, constant schedule
Batching and clipping	Microbatch 1, accumulation 4, end-of-epoch flush, norm 1.0
Targets	Prompt masked; leading space, stripped solution, newline, supervised EOS
Training	Two epochs, max 1024 tokens, FP16 GradScaler, final adapter
Memory settings	Gradient checkpointing, use_reentrant=False, cache disabled

Appendix B Dated decision rules

The following quotations retain the relevant decision language from the dated files. The original pilot on instances 0–4 had no dated efficacy rule and is exploratory. The expanded pilot used new instances from previously selected families. Zero endpoints do not count as excluding zero.

B1 Expanded pilot evaluation on October 12 2026

“Prespecified nominal evidence of benefit: the lower endpoint of the paired original-family percentile bootstrap 95% interval for this primary difference is strictly greater than zero (10,000 replicates; bootstrap seed 2026). An interval containing zero is inconclusive, regardless of a positive point estimate. An upper endpoint below zero is evidence of harm under this criterion.”

“Call a nominal benefit robust to formatting only if the numeric-scoring primary interval also has a strictly positive lower endpoint. Otherwise report the discrepancy explicitly.”

Primary difference: equal three-seed pooled pilot-minus-base accuracy on instances 5–14. Source: gsm_symbolic_success_rule_2026-10-12.txt.

B2 Difficulty targeting on October 12 2026

“Hard wins only if the entire primary 95% interval is above zero. An interval entirely below zero favors random. Otherwise the result is inconclusive; an endpoint exactly equal to zero does not exclude zero.”

“Either direction excluding zero meets the previously agreed gate for considering the secondary, existing 300-example training comparison.” That gate did not itself authorize another experiment. Source: rq2_analysis_plan_2026-10-12.md.

B3 Human reference control on October 13 2026

“The gold control "works" only if the entire paired 95% family-bootstrap interval is above zero. An interval wholly below zero indicates worse performance; otherwise the control is inconclusive. An endpoint exactly equal to zero does not pass. Treat a borderline interval cautiously.”

The primary contrast is the equal three-seed reference-minus-base mean. The plan separately specifies “Secondary: gold minus hard, comparing three-seed means with the same shared family draws.” It assigns no separate efficacy gate to that secondary contrast. Source: gold_control_plan_2026-10-13.md.

B4 Scaled random training on October 13 2026

“Success requires BOTH the pooled family-only 95% bootstrap interval and the pooled seed-and-family 95% bootstrap interval to have lower endpoints strictly above zero.”

“If only family-only passes: report ‘family-only pass, not robust to seed variation’, not success.” A seed-and-family-only pass also fails, and zero endpoints fail. The amended plan supersedes the draft rule. Sources: scaled_random_self_training_plan_2026-10-13.md and scaled_random_self_training_amended_plan_2026-10-13.md.

Appendix C Accuracy by training seed

Table C1 Accuracy percentages on the common 400 question evaluation

Condition	Seed	P1 n 200	P2 n 200	Pooled n 400
Base	—	49.00	29.50	39.25
Pilot 43	2026	51.00	32.00	41.50
Pilot 43	2027	52.50	33.50	43.00
Pilot 43	2028	53.00	34.00	43.50
Hard 300	2026	51.00	30.00	40.50
Hard 300	2027	52.00	32.00	42.00
Hard 300	2028	53.00	29.50	41.25
Random 353	2026	54.00	31.50	42.75
Random 353	2027	54.00	29.50	41.75
Random 353	2028	53.00	31.00	42.00
Easy 403	2026	54.50	25.00	39.75
Easy 403	2027	48.00	24.00	36.00
Easy 403	2028	53.50	25.00	39.25
References 300	2026	35.50	22.00	28.75
References 300	2027	42.00	26.50	34.25
References 300	2028	46.50	25.00	35.75
Scaled 2000	2026	57.50	29.50	43.50
Scaled 2000	2027	51.50	30.00	40.75
Scaled 2000	2028	50.00	26.50	38.25

All trained groups have three seeds. Group means average seed accuracies equally. The baseline is one fixed model, reused across comparisons. There are 400 distinct evaluation questions, not 1,200 independent questions per condition.

Table C2 Exploratory pilot on instances 0 to 4

Model	P1 accuracy	P2 accuracy	Pooled accuracy
Base	51.00	24.00	37.50
Pilot 2026	54.00	26.00	40.00
Pilot 2027	56.00	25.00	40.50
Pilot 2028	53.00	26.00	39.50
Pilot mean	54.33	25.67	40.00

The exploratory set has 100 questions per variant, 200 total. The mean gain is 2.50 percentage points. It is separate from the later 400-question evaluation and has no located prospective success rule.

Appendix D All variant level contrasts

Table D1 reports every P1/P2 contrast corresponding to Table 2. Family-only secondary intervals are unadjusted. All variant-level seed-and-family intervals in this table are post hoc sensitivities; the prospectively specified seed-and-family analyses concerned pooled scaled contrasts. Variant results do not replace a pooled primary criterion.

Table D1 Variant contrasts in percentage points

Comparison	Set	Change	95% family interval	95% seed and family interval
Pilot 43 − Base	P1	+3.17	[−2.94, +9.30]	[−2.98, +9.41]
Pilot 43 − Base	P2	+3.67	[−2.78, +10.22]	[−2.83, +10.50]
Hard 300 − Random 353	P1	−1.67	[−6.84, +3.49]	[−7.83, +4.07]
Hard 300 − Random 353	P2	−0.17	[−7.02, +5.56]	[−8.33, +6.19]
References 300 − Base	P1	−7.67	[−19.84, +5.24]	[−20.74, +6.67]
References 300 − Base	P2	−5.00	[−13.91, +3.89]	[−14.39, +4.44]
Scaled 2000 − Base	P1	+4.00	[−1.05, +10.00]	[−2.00, +11.50]
Scaled 2000 − Base	P2	−0.83	[−6.08, +4.17]	[−6.51, +4.91]
Hard 300 − Base	P1	+3.00	[−2.63, +8.67]	[−2.86, +9.17]
Hard 300 − Base	P2	+1.00	[−5.37, +7.89]	[−5.78, +8.15]
Random 353 − Base	P1	+4.67	[+0.14, +9.63]	[−0.37, +10.00]
Random 353 − Base	P2	+1.17	[−5.08, +8.04]	[−5.33, +8.33]
Easy 403 − Base	P1	+3.00	[−4.71, +10.79]	[−5.56, +11.58]
Easy 403 − Base	P2	−4.83	[−11.45, +0.73]	[−11.83, +1.05]
Hard 300 − Easy 403	P1	0.00	[−5.97, +5.93]	[−7.45, +7.33]
Hard 300 − Easy 403	P2	+5.83	[0.00, +12.67]	[−0.42, +13.67]
Random 353 − Easy 403	P1	+1.67	[−5.26, +8.57]	[−6.30, +10.00]
Random 353 − Easy 403	P2	+6.00	[−1.52, +17.00]	[−1.78, +17.07]
References 300 − Hard 300	P1	−10.67	[−23.17, +1.50]	[−23.50, +3.33]
References 300 − Hard 300	P2	−6.00	[−14.76, +2.00]	[−15.61, +2.64]
Scaled 2000 − Random 353	P1	−0.67	[−4.70, +4.45]	[−6.90, +6.09]
Scaled 2000 − Random 353	P2	−2.00	[−8.55, +4.26]	[−9.68, +4.58]

The random-minus-base P1 family interval narrowly excludes zero; the P2 hard-minus-easy lower endpoint equals zero. Individual seed scores are descriptive; for example, scaled seed 2026 has a P1 gain of 8.50 points with a family interval [+1.05, +16.43]. The complete seed table is reported in Appendix C, while the experiment’s primary decision uses the three-seed pooled contrast.

Appendix E Response diagnostics

Each row below contains 400 saved responses to the same questions. Cap means exactly 384 generated tokens. Unparsed means a null saved prediction. “Same” counts identical non-null saved prediction strings relative to base; it includes wrong answers that agree and excludes both-null pairs. All rates in Figure 2 divide by the full response count.

Table E1 Completion and parsed answer agreement

Condition and seed	Mean tokens	Cap	Unparsed	Cap and unparsed	Same
Base	174.32	20	12	12	—
Pilot 43 2026	165.46	4	2	1	193
Pilot 43 2027	164.83	4	2	2	194
Pilot 43 2028	166.81	4	1	1	196
Hard 300 2026	170.12	11	11	10	185
Hard 300 2027	167.96	14	13	13	191
Hard 300 2028	164.23	10	8	8	183
Random 353 2026	163.58	5	6	5	199
Random 353 2027	162.89	7	8	7	195
Random 353 2028	161.98	3	4	3	195
Easy 403 2026	156.89	5	5	5	178
Easy 403 2027	159.88	7	8	7	164
Easy 403 2028	161.85	5	5	5	167
References 300 2026	176.79	20	20	20	95
References 300 2027	171.97	14	13	13	111
References 300 2028	174.72	18	16	16	115
Scaled 2000 2026	163.39	8	8	7	197
Scaled 2000 2027	166.51	12	11	11	193
Scaled 2000 2028	168.94	10	10	10	198

Pooled hard: 35 cap hits, 32 unparsed answers, and 31 outputs with both, across 1,200 responses. Pooled references: 52 cap hits, 49 unparsed answers, and 49 outputs with both, across 1,200 responses. Base: 20, 12, and 12 respectively across 400 responses. Pooled reference-versus-base answer agreement is 321/1,200 (26.75%); hard/random/easy combined agreement is 1,657/3,600 (46.03%).

Across the 12 hard/random/easy/scaled adapters, minimum and maximum verdict flips are 54 and 79, and net changes are −13 to +17. Pilot flips are 75, 79, and 83. Reference flips are 110, 104, and 106. Fixes and regressions are paired comparisons with the same base question, not additional independent evaluations.

Appendix F Scoring and failure records

F1 Scoring checks

The legacy extractor accepts a leading numeric fragment after #### and removes commas. Saved examples include “75,35” parsed as 7535 and “10G / 3” parsed as 10. An audit found 86 distinct non-clean suffixes across the pre-scaled outputs, one scored correct. In that case a suffix beginning “80%” continued into another generated question, while the saved prediction 80 matched the reference. The numeric-equality audit concerns equality after extraction and does not validate the extraction itself.

The approved stricter scorer was applied as a secondary diagnostic to newly generated scaled responses, without changing the original scoring rule. It changed one of 1,200 scaled verdicts: a “4/45” answer had been parsed as 4 by the legacy scorer. Numeric-equality checks changed no expanded-pilot or RQ2 verdicts. The primary tables preserve the saved legacy scores.

F2 Targeted reasoning review

Shuffled position	Recorded finding
1221	Two cancelling errors: outside-cleaning multiplier and number of workers.
613	Two cancelling errors: discount interpreted as a price and savings subtraction reversed.
807	Flawed but repairable bookkeeping explanation; borderline qualification retained.

The review selected 20 hard solutions, emphasizing problems with only one correct attempt. The author confirmed these three flags on October 14, 2026; no targets were removed. The sample is not an estimate of the population reasoning-error rate.

F3 Operational failures and provenance

Record	Outcome
Initial reference launch	A CUDA-disabling check prevented training. The corrected run supplied the reported reference models.
Interrupted batch size run	The incomplete batch-4 check was preserved; a complete rerun supplied the reported 38.25% result.
CPU preflight repairs	Identity/hash receipt and test-wrapper issues were repaired before the corresponding scaled stages; the training engine was retained.
Memory counters	Some framework reserved-memory counters exceeded nominal VRAM. Completed runs do not establish a cause or system-RAM spill.
Pilot provenance	No original model commit, supervised-token totals, or reliable successful/skipped update accounting was recorded.

All 11 pooled family-only intervals were independently reproduced from saved verdicts to numerical tolerance of 1e-10. The saved reference and scaled seed-and-family intervals were also reproduced. Additional seed sensitivities use the same paired sampling method, with their post hoc status recorded in Tables 2 and D1.